

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

I M. Pranay (192525382) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

M. Pranay

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s=255-r$ to the pixel values:

$r = [50, 100, 150, 200]$

Compute the transformed values.

A.

(A) Grey level transformation is used to modify the intensity values of an image to improve its appearance. It helps to:

- Enhance image contrast.
- Brighten or darken an image.
- Highlight important features.
- Prepare images for further processing such as segmentation and object detection.

(B) Formula:

$$s = 255 - r$$

Given:

$$r = [50, 100, 150, 200]$$

Calculation:

Original Pixel (r)	Transformed Pixel (s = 255-r)
50	205
100	155
150	105
200	55

Answer:

$$[205, 155, 105, 55]$$

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s=c\log(1+r)$, where $c=10$, compute s for:

$r = [0, 10, 50, 100]$

A.

(a) Log transformation expands the values of dark pixels while compressing the values of bright pixels. It improves the visibility of details in darker regions without greatly affecting brighter regions.

Formula:

$$s=c\log(1+r)$$

where

c = constant

r = input pixel value

(b) **Given**

$$c = 10$$

Using natural logarithm (ln)

$$s = 10\ln(1 + r)$$

r	Calculation	s
0	$10 \ln(1)$	0
10	$10 \ln(11)$	23.98
50	$10 \ln(51)$	39.32
100	$10 \ln(101)$	46.15

Answer:

[0, 23.98, 39.32, 46.15]

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

A.

(a) Histogram equalization improves image contrast by redistributing the intensity values uniformly across the available range. It enhances details in both dark and bright regions and produces a clearer image.

(b) **Given Histogram**

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels

$$N = 2 + 2 + 4 + 8 = 16$$

Step 1: Cumulative Frequency

Intensity	Frequency	Cumulative Frequency
0	2	2
1	2	4

Intensity	Frequency	Cumulative Frequency
2	4	8
3	8	16

Step 2: Cumulative Distribution Function (CDF)

CDF = Cumulative Frequency / Total Pixels

Intensity	CDF
0	2/16=0.125
1	4/16=0.25
2	8/16=0.50
3	16/16=1.00

Step 3: Equalized Intensities

For a **3-bit image**, number of gray levels:

$$L = 8$$

Formula

$$s = (L - 1) \times CDF$$

$$s = 7 \times CDF$$

Intensity	CDF	Equalized Value
0	0.125	1
1	0.25	2
2	0.50	4
3	1.00	7

Answer

Original Intensity	Equalized Intensity
0	1
1	2
2	4
3	7

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

A.

(a) Spatial smoothing reduces image noise and unwanted details by replacing each pixel with the average of its neighboring pixels. It produces a smoother image before segmentation or feature extraction.

(c) Apply a 3×3 averaging filter

Matrix

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$$

Average Filter

Sum

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Average

$$\frac{270}{9} = 30$$

New center pixel

$$\boxed{30}$$

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$$[0 \ -1 \ 0], \{-1 \ 4 \ -1\}, \{0 \ -1 \ 0\}$$

to the matrix:

$$10 \ 10 \ 10, 10 \ 20 \ 10, 10 \ 10 \ 10$$

Compute the output at the center.

A.

(a) Sharpening increases the contrast around edges, making boundaries and fine details more visible. It is commonly used after smoothing to improve image clarity.

(b) Apply Laplacian Mask

Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Calculation

$$\begin{aligned} &= (4 \times 20) - (10 + 10 + 10 + 10) \\ &= 80 - 40 \\ &= 40 \end{aligned}$$

Output at center

$$\boxed{40}$$